RISC (Reduced Instruction Set Computing) based designs require less transistors than the conventional CISC (Complex Instruction Set Computing) based designs that power most of the personal computers. They are more power efficient and compact making them the ideal choice for smartphones, tablets and other such embedded systems.

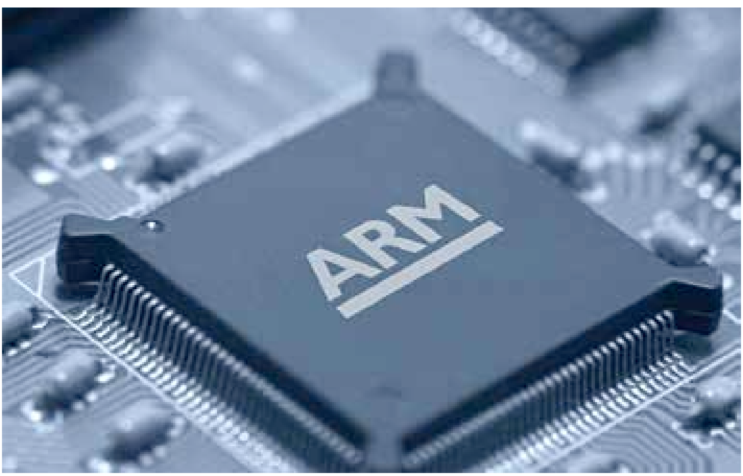
More than 100 billion ARM processors have been produced up to 2017 making ARM the most popular ISA (Instruction Set Architecture) in terms of number.

The ARM architecture

The latest generation of ARM architecture is the ARMv8-A. A significant improvement over the last version, the v8 adds an optional 64-bit architecture to it.

The latest version of processors based on this architecture come with a number of advancements and added features including:

- Enhanced memory models increasing the efficiency of the chip
- Architecture based on big.LITTLE configuration allowing two different cores with different clock speed to work together on a single processor
- New instruction set called AArch64 that comes with a new exception system, Advanced SIMD enhancements and a 48-bit virtual address space based on LPAE (Large Physical Address Extension) capable of extending it to 64-bit



ARM processors dominate the mobile SoC market currently.